QUANT APTITUDE COACH

Verified worked solutions · 21 checked practice questions

Answer keys are cross-checked with the source answer list on PDF page 23. Each solution below is independently recomputed. Source question references point to the user-held PDF pages 14-15.

Question 1 · Numbers · Beginner

The difference between the local value and face value of 7 in the numeral 657903 is:

(A) 0

(B) 7896

(C) 6993

(D) 903

CORRECT ANSWER: C · 6993

WORKED SOLUTION

The digit 7 is in the thousands place, so its place (local) value is 7,000. Its face value is 7. Difference = 7,000 - 7 = 6,993.

Source question: original PDF page 14, Exercise 1, Question 1. Answer-key cross-check: original PDF page 23.

Question 2 · Numbers · Beginner

The difference between the place values of 7 and 3 in the number 527435 is:

(A) 4

(B) 5

(C) 45

(D) 6970

CORRECT ANSWER: D · 6970

WORKED SOLUTION

In 527435, the place value of 7 is 7,000 and the place value of 3 is 30. Difference = 7,000 - 30 = 6,970.

Source question: original PDF page 14, Exercise 1, Question 2. Answer-key cross-check: original PDF page 23.

Question 3 · Numbers · Beginner

The sum of the smallest six-digit number and the greatest five-digit number is:

(A) 199999

(B) 201110

(C) 211110

(D) 1099999

CORRECT ANSWER: A · 199999

WORKED SOLUTION

The smallest six-digit number is 100,000. The greatest five-digit number is 99,999. Sum = 100,000 + 99,999 = 199,999.

Source question: original PDF page 14, Exercise 1, Question 3. Answer-key cross-check: original PDF page 23.

Question 4 · Numbers · Beginner

If the largest three-digit number is subtracted from the smallest five-digit number, the remainder is:

(A) 1

(B) 9000

(C) 9001

(D) 90001

CORRECT ANSWER: C · 9001

WORKED SOLUTION

The smallest five-digit number is 10,000 and the largest three-digit number is 999. Remainder = 10,000 - 999 = 9,001.

Source question: original PDF page 14, Exercise 1, Question 4. Answer-key cross-check: original PDF page 23.

Question 5 · Numbers · Beginner

5978 + 6134 + 7014 = ?

(A) 16226

(B) 19126

(C) 19216

(D) 19226

CORRECT ANSWER: B · 19126

WORKED SOLUTION

Add in steps: 5,978 + 6,134 = 12,112. Then 12,112 + 7,014 = 19,126.

Source question: original PDF page 14, Exercise 1, Question 5. Answer-key cross-check: original PDF page 23.

Question 7 · Numbers · Beginner

39798 + 3798 + 378 = ?

(A) 43576

(B) 43974

(C) 43984

(D) 49532

CORRECT ANSWER: B · 43974

WORKED SOLUTION

Add 3,798 + 378 = 4,176. Then 39,798 + 4,176 = 43,974.

Source question: original PDF page 14, Exercise 1, Question 7. Answer-key cross-check: original PDF page 23.

Question 8 · Numbers · Beginner

9358 - 6014 + 3127 = ?

(A) 6381

(B) 6471

(C) 6561

(D) 6741

CORRECT ANSWER: B · 6471

WORKED SOLUTION

Evaluate from left to right: 9,358 - 6,014 = 3,344. Then 3,344 + 3,127 = 6,471.

Source question: original PDF page 14, Exercise 1, Question 8. Answer-key cross-check: original PDF page 23.

Question 9 · Numbers · Intermediate

9572 - 4018 - 2164 = ?

(A) 3300

(B) 3390

(C) 3570

(D) 7718

CORRECT ANSWER: B · 3390

WORKED SOLUTION

Evaluate from left to right: 9,572 - 4,018 = 5,554. Then 5,554 - 2,164 = 3,390.

Source question: original PDF page 14, Exercise 1, Question 9. Answer-key cross-check: original PDF page 23.

Question 10 · Numbers · Intermediate

7589 - ? = 3434. What number replaces the question mark?

(A) 721

(B) 3246

(C) 4155

(D) 11028

CORRECT ANSWER: C · 4155

WORKED SOLUTION

Let the missing number be x. From 7,589 - x = 3,434, x = 7,589 - 3,434 = 4,155.

Source question: original PDF page 14, Exercise 1, Question 10. Answer-key cross-check: original PDF page 23.

Question 12 · Numbers · Intermediate

7545 - ? = 8461 - 3569. What number replaces the question mark?

(A) 2593

(B) 2773

(C) 3569

(D) None of these

CORRECT ANSWER: D · None of these

WORKED SOLUTION

First evaluate the right-hand side: 8,461 - 3,569 = 4,892. Therefore the missing number is 7,545 - 4,892 = 2,653. Since 2,653 is not listed, the answer is None of these.

Source question: original PDF page 14, Exercise 1, Question 12. Answer-key cross-check: original PDF page 23.

Question 13 · Numbers · Intermediate

3578 + 5729 - ?486 = 5821. Which digit replaces the question mark?

(A) 1

(B) 2

(C) 3

(D) None of these

CORRECT ANSWER: C · 3

WORKED SOLUTION

First add 3,578 + 5,729 = 9,307. The number being subtracted must be 9,307 - 5,821 = 3,486. Thus the missing leading digit is 3.

Source question: original PDF page 14, Exercise 1, Question 13. Answer-key cross-check: original PDF page 23.

Question 17 · Numbers · Intermediate

5358 x 51 = ?

(A) 273258

(B) 273268

(C) 273348

(D) 273358

CORRECT ANSWER: A · 273258

WORKED SOLUTION

5358 x 51 = 5358 x (50 + 1) = 267,900 + 5,358 = 273,258.

Source question: original PDF page 15, Exercise 1, Question 17. Answer-key cross-check: original PDF page 23.

Question 18 · Numbers · Beginner

360 x 17 = ?

(A) 5120

(B) 5320

(C) 6120

(D) 6130

CORRECT ANSWER: C · 6120

WORKED SOLUTION

360 x 17 = 360 x (10 + 7) = 3,600 + 2,520 = 6,120.

Source question: original PDF page 15, Exercise 1, Question 18. Answer-key cross-check: original PDF page 23.

Question 19 · Numbers · Intermediate

587 x 999 = ?

(A) 586413

(B) 587523

(C) 614823

(D) 615173

CORRECT ANSWER: A · 586413

WORKED SOLUTION

Use 999 = 1,000 - 1. So 587 x 999 = 587 x 1,000 - 587 = 587,000 - 587 = 586,413.

Source question: original PDF page 15, Exercise 1, Question 19. Answer-key cross-check: original PDF page 23.

Question 25 · Numbers · Intermediate

1014 x 986 = ?

(A) 998804

(B) 998814

(C) 998904

(D) 999804

CORRECT ANSWER: D · 999804

WORKED SOLUTION

Write the factors as (1,000 + 14) and (1,000 - 14). Using $(a+b)(a-b)=a^2-b^2$: $1{,}000^2 - 14^2 = 1{,}000{,}000 - 196 = 999{,}804$.

Source question: original PDF page 15, Exercise 1, Question 25. Answer-key cross-check: original PDF page 23.

Question 26 · Numbers · Pro

1307 x 1307 = ?

(A) 1601249

(B) 1607249

(C) 1701249

(D) 1708249

CORRECT ANSWER: D · 1708249

WORKED SOLUTION

$1307^2 = (1{,}300 + 7)^2 = 1{,}300^2 + 2 \times 1{,}300 \times 7 + 7^2 = 1{,}690{,}000 + 18{,}200 + 49 = 1{,}708{,}249$.

Source question: original PDF page 15, Exercise 1, Question 26. Answer-key cross-check: original PDF page 23.

Question 27 · Numbers · Pro

1399 x 1399 = ?

(A) 1687401

(B) 1901541

(C) 1943211

(D) 1957201

CORRECT ANSWER: D · 1957201

WORKED SOLUTION

Use 1,399 = 1,400 - 1. Then $1{,}399^2 = 1{,}400^2 - 2 \times 1{,}400 \times 1 + 1^2 = 1{,}960{,}000 - 2{,}800 + 1 = 1{,}957{,}201$.

Source question: original PDF page 15, Exercise 1, Question 27. Answer-key cross-check: original PDF page 23.

Question 28 · Numbers · Pro

106 x 106 + 94 x 94 = ?

(A) 20032

(B) 20072

(C) 21032

(D) 23032

CORRECT ANSWER: B · 20072

WORKED SOLUTION

$106^2 = 11{,}236$ and $94^2 = 8{,}836$. Their sum is $11{,}236 + 8{,}836 = 20{,}072$. Alternatively use $(100+6)^2 + (100-6)^2 = 2(100^2+6^2)$.

Source question: original PDF page 15, Exercise 1, Question 28. Answer-key cross-check: original PDF page 23.

Question 29 · Numbers · Pro

217 x 217 + 183 x 183 = ?

(A) 79698

(B) 80578

(C) 80698

(D) 81268

CORRECT ANSWER: B · 80578

WORKED SOLUTION

Write 217 = 200 + 17 and 183 = 200 - 17. Using $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$: $2(200^2 + 17^2) = 2(40{,}000 + 289) = 80{,}578$.

Source question: original PDF page 15, Exercise 1, Question 29. Answer-key cross-check: original PDF page 23.

Question 31 · Numbers · Pro

What number should replace x in this multiplication problem? 3x4 x 4 = 1216.

(A) 0

(B) 2

(C) 4

(D) 5

CORRECT ANSWER: A · 0

WORKED SOLUTION

Divide the product by 4: 1,216 / 4 = 304. Therefore the three-digit number 3x4 is 304, so x = 0.

Source question: original PDF page 15, Exercise 1, Question 31. Answer-key cross-check: original PDF page 23.

Question 32 · Numbers · Pro

A positive integer, when added to 1000, gives a sum greater than when it is multiplied by 1000. This positive integer is:

(A) 1

(B) 3

(C) 5

(D) 7

CORRECT ANSWER: A · 1

WORKED SOLUTION

Let the positive integer be n. The condition is n + 1,000 > 1,000n, so 1,000 > 999n and n < 1,000/999, which is just above 1. The only positive integer satisfying this is n = 1.

Source question: original PDF page 15, Exercise 1, Question 32. Answer-key cross-check: original PDF page 23.